Rigorous Gradient Derivation for the Ω-Framework

With First-Order Taylor Decomposition and
Comparison to Giannou et al. (2022)

Eugene Shcherbinin

March 2026

Contents

1 Setup and Notation

1.1 The Data Pipeline

Fix the following objects from the Ω-framework (Items 1–19):

Symbol	Space	Description
$\mathcal{X}$	$\in O$	Observed reality (raw experience)
$X = \tau(\mathcal{X})$	$\in \Pi = W \cup S$	Proposition (rationalised observation)
$\mathbf{X} = \phi_\psi(X)$	$\in \mathbb{R}^m$	Embedding (numerical representation)
$Y = f(\mathbf{X})$	$\in \mathbb{R}^k$	True causal map (unknown)
$\hat{Y} = \hat{f}_\theta(\mathbf{X})$	$\in \mathbb{R}^k$	Learned approximation
$\mathcal{L}(Y, \hat{Y})$	$\in \mathbb{R}$	Loss function
θ	$\in \mathbb{R}^l$	Model parameters
ψ	$\in \mathbb{R}^p$	Embedding parameters
$\beta = (\theta, \psi)$	$\in \mathbb{R}^{l+p}$	All learnable parameters

1.2 Assumptions

(A1) $\mathcal{L} : \mathbb{R}^k \times \mathbb{R}^k \to \mathbb{R}$ is continuously differentiable in both arguments.

(A2) $\hat{f}_\theta : \mathbb{R}^m \to \mathbb{R}^k$ is continuously differentiable in θ and in its input.

(A3) $\phi_\psi : \Pi \to \mathbb{R}^m$ is continuously differentiable in ψ (via UPRT, Item 9).

(A4) $f : \mathbb{R}^m \to \mathbb{R}^k$ is the true causal map, continuously differentiable (when accessible).

1.3 Dimension Summary

For clarity, we annotate every Jacobian with its dimensions:

$$\frac{\partial \mathcal{L}}{\partial Y} \in \mathbb{R}^{1\times k}, \qquad \frac{\partial \mathcal{L}}{\partial \hat{Y}} \in \mathbb{R}^{1\times k},$$
$$\frac{\partial f}{\partial \mathbf{X}} \in \mathbb{R}^{k\times m}, \qquad \frac{\partial \hat{f}_\theta}{\partial \mathbf{X}} \in \mathbb{R}^{k\times m},$$
$$\frac{\partial \hat{f}_\theta}{\partial \theta} \in \mathbb{R}^{k\times l}, \qquad \frac{\partial \phi_\psi}{\partial \psi} \in \mathbb{R}^{m\times p}.$$

2 The Gradient $\nabla_\beta \log \mathcal{L}$

2.1 Composite Structure

The loss is:

$$\mathcal{L} = \mathcal{L}\big(f(\phi_\psi(X)),\ \hat{f}_\theta(\phi_\psi(X))\big). \tag{1}$$

Write $\mathbf{X}_\psi := \phi_\psi(X)$ for the embedding. Then $Y = f(\mathbf{X}_\psi)$ and $\hat{Y} = \hat{f}_\theta(\mathbf{X}_\psi)$, so:

$$\mathcal{L} = \mathcal{L}(Y(\psi), \hat{Y}(\theta, \psi)).$$

Note: Y depends on ψ (through the embedding) but *not* on θ. $\hat{Y}$ depends on both.

2.2 Gradient of $\log \mathcal{L}$

By the chain rule for log:

$$\boxed{\nabla_\beta \log \mathcal{L} = \frac{1}{\mathcal{L}} \cdot \nabla_\beta \mathcal{L}} \tag{2}$$

So we need $\nabla_\theta \mathcal{L}$ and $\nabla_\psi \mathcal{L}$.

2.3 Gradient w.r.t. θ (the model parameters)

Since $Y = f(\mathbf{X}_\psi)$ does not depend on θ:

$$\boxed{\nabla_\theta \mathcal{L} = \underbrace{\frac{\partial \mathcal{L}}{\partial \hat{Y}}}_{1\times k} \cdot \underbrace{\frac{\partial \hat{f}_\theta(\mathbf{X}_\psi)}{\partial \theta}}_{k\times l} \in \mathbb{R}^{1\times l}} \tag{3}$$

This is **standard backpropagation** (Item 19, eq. 1).
Therefore:

$$\nabla_\theta \log \mathcal{L} = \frac{1}{\mathcal{L}} \cdot \frac{\partial \mathcal{L}}{\partial \hat{Y}} \cdot \frac{\partial \hat{f}_\theta}{\partial \theta}. \tag{4}$$

2.4 Gradient w.r.t. ψ (the embedding parameters)

Both arguments of $\mathcal{L}$ depend on ψ through $\mathbf{X}_\psi = \phi_\psi(X)$. By the multivariate chain rule:

$$\boxed{\nabla_\psi \mathcal{L} = \left[\underbrace{\frac{\partial \mathcal{L}}{\partial Y} \cdot \frac{\partial f}{\partial \mathbf{X}}}_{\text{through } f \text{ (causal)}} + \underbrace{\frac{\partial \mathcal{L}}{\partial \hat{Y}} \cdot \frac{\partial \hat{f}_\theta}{\partial \mathbf{X}}}_{\text{through } \hat{f}_\theta \text{ (learned)}} \right] \cdot \frac{\partial \phi_\psi}{\partial \psi} \in \mathbb{R}^{1\times p}} \tag{5}$$

The two-term structure (Item 19, eq. 2):

- **Term A** (causal): $\frac{\partial \mathcal{L}}{\partial Y} \cdot \frac{\partial f}{\partial \mathbf{X}} \cdot \frac{\partial \phi_\psi}{\partial \psi}$ — propagates through the *true* causal map f. **Inaccessible** if f is unknown.
- **Term B** (learned): $\frac{\partial \mathcal{L}}{\partial \hat{Y}} \cdot \frac{\partial \hat{f}_\theta}{\partial \mathbf{X}} \cdot \frac{\partial \phi_\psi}{\partial \psi}$ — propagates through the *learned* approximation $\hat{f}_\theta$. Always computable.

Therefore:

$$\nabla_\psi \log \mathcal{L} = \frac{1}{\mathcal{L}} \left[\frac{\partial \mathcal{L}}{\partial Y} \cdot \frac{\partial f}{\partial \mathbf{X}} + \frac{\partial \mathcal{L}}{\partial \hat{Y}} \cdot \frac{\partial \hat{f}_\theta}{\partial \mathbf{X}} \right] \cdot \frac{\partial \phi_\psi}{\partial \psi}. \tag{6}$$

2.5 The Full Gradient $\nabla_\beta \log \mathcal{L}$

Combining (**??**) and (**??**):

Key Result

Let $\beta = (\theta, \psi) \in \mathbb{R}^{l+p}$. Then:

$$\nabla_\beta \log \mathcal{L} = \frac{1}{\mathcal{L}} \begin{pmatrix} \dfrac{\partial \mathcal{L}}{\partial \hat{Y}} \cdot \dfrac{\partial \hat{f}_\theta}{\partial \theta} \\ \left[\dfrac{\partial \mathcal{L}}{\partial Y} \cdot \dfrac{\partial f}{\partial \mathbf{X}} + \dfrac{\partial \mathcal{L}}{\partial \hat{Y}} \cdot \dfrac{\partial \hat{f}_\theta}{\partial \mathbf{X}} \right] \cdot \dfrac{\partial \phi_\psi}{\partial \psi} \end{pmatrix} \tag{7}$$

The θ-block has dimension l. The ψ-block has dimension p. The ψ-block contains the inaccessible term $\partial f / \partial \mathbf{X}$.

3 First-Order Taylor Expansion and the Trinity Decomposition

3.1 Taylor Expansion of the Loss

Let $Y^* \in \mathbb{R}^k$ be a reference prediction (e.g., the Bayes-optimal predictor at a stationary θ^*). Define the residual $\delta = \hat{Y} - Y^*$. Taylor-expanding $\mathcal{L}$ in its second argument:

$$\mathcal{L}(Y, Y^* + \delta) = \mathcal{L}(Y, Y^*) + \nabla_{\hat{Y}} \mathcal{L}(Y, Y^*)^\top \delta + \frac{1}{2} \delta^\top \nabla^2_{\hat{Y}} \mathcal{L}(Y, Y^*)\, \delta + \mathcal{O}(\|\delta\|^3). \tag{8}$$

3.2 The Expected Risk and Trinity Decomposition

The expected risk (Item 20):

$$R(\theta, \psi) = \mathbb{E}_{\mathcal{X} \sim p}\big[\mathcal{L}\big(f(\phi_\psi(\tau(\mathcal{X}))),\ \hat{f}_\theta(\phi_\psi(\tau(\mathcal{X})))\big)\big]. \tag{9}$$

By the Trinity partition (Item 21), since $\tau : O \to \Pi$ is a bijection and $\Pi = W \cup S$:

$$\boxed{R(\theta, \psi) = P_W \cdot \mathbb{E}[\mathcal{L} \mid X \in W] + P_S \cdot \mathbb{E}[\mathcal{L} \mid X \in S]} \tag{10}$$

where $P_W = P_\theta(\tau^{-1}(W))$ and $P_S = P_\theta(\tau^{-1}(S))$.

3.3 Taylor + Trinity = Loss Decomposition

Substituting (**??**) into (**??**) and writing $H_{\hat{Y}} = \nabla^2_{\hat{Y}} \mathcal{L}(Y, Y^*)$:

Key Result

$$\begin{aligned} R(\theta, \psi) \approx\ & \underbrace{P_W \cdot \mathbb{E}[\mathcal{L}(Y, Y^*) \mid W] + P_S \cdot \mathbb{E}[\mathcal{L}(Y, Y^*) \mid S]}_{\text{irreducible loss at reference}} \\ & + \underbrace{P_W \cdot \mathbb{E}[\nabla_{\hat{Y}} \mathcal{L}^\top \delta \mid W] + P_S \cdot \mathbb{E}[\nabla_{\hat{Y}} \mathcal{L}^\top \delta \mid S]}_{\text{first-order correction (bias)}} \\ & + \underbrace{\tfrac{1}{2} P_W \cdot \mathbb{E}[\delta^\top H_{\hat{Y}} \delta \mid W] + \tfrac{1}{2} P_S \cdot \mathbb{E}[\delta^\top H_{\hat{Y}} \delta \mid S]}_{\text{second-order correction (variance / curvature)}} \end{aligned} \tag{11}$$

Interpretation (this is the key entry point):

1. **Row 1**: Irreducible error — the loss you'd get even with the best possible predictor. Includes Pirsig's "Quality loss" (information destroyed by τ).
2. **Row 2**: Bias — directional misalignment of $\hat{Y}$ from Y^*. Vanishes at a stationary point. Decomposes separately over W and S: the learner may be biased differently on truths vs. falsehoods.
3. **Row 3**: Variance/curvature — the cost of dispersion around Y^*. The Hessian $H_{\hat{Y}}$ controls how sensitive the loss is to prediction errors. Also decomposes over W and S.

3.4 Gradient of the Trinity-Expanded Risk

Differentiating (**??**) w.r.t. θ, and noting that P_W, P_S may depend on θ if the sampling distribution p_θ does:

$$\nabla_\theta R = P_W\, \mathbb{E}[\mathcal{L} \cdot s_\theta + \nabla_\theta \mathcal{L} \mid W] + P_S\, \mathbb{E}[\mathcal{L} \cdot s_\theta + \nabla_\theta \mathcal{L} \mid S] \tag{12}$$

where $s_\theta(\mathcal{X}) := \nabla_\theta \log p_\theta(\mathcal{X})$ is the **score function**.

Each conditional expectation contains both the REINFORCE term ($\mathcal{L} \cdot s_\theta$, exploration) and the backprop term ($\nabla_\theta \mathcal{L}$, exploitation). The Trinity splits each into a W-component and an S-component.

4 Parameter-Space Taylor Expansion

Alternatively, expand R in θ around a stationary point θ^* where $\nabla_\theta R(\theta^*) = 0$. Let $\delta_\theta = \theta - \theta^*$:

$$R(\theta^* + \delta_\theta) \approx R(\theta^*) + \frac{1}{2}\delta_\theta^\top \Big(\underbrace{H_W^\theta}_{W\text{-curvature}} + \underbrace{H_S^\theta}_{S\text{-curvature}} \Big)\delta_\theta \tag{13}$$

where $H_A^\theta = \nabla_\theta^2 R_A(\theta^*)$ for $A \in \{W, S\}$. The eigenspectra of H_W^θ and H_S^θ reveal whether the learner's parameter sensitivity comes from getting truths wrong or from getting falsehoods wrong.

5 Comparison with Giannou et al. (2022)

Giannou et al. study the convergence of policy gradient methods to Nash equilibria in general stochastic games. Their framework:

- N-player stochastic game $\mathcal{G} = (\mathcal{S}, \mathcal{N}, \{\mathcal{A}_i\}, \{R_i\}, P, \zeta, \rho)$
- Policy gradient update: $\pi_{n+1} = \mathrm{proj}_{\Pi}(\pi_n + \gamma_n \hat{v}_n)$
- Gradient dominance property (GDP): $V_{i,\rho}(\pi_i'; \pi_{-i}) - V_{i,\rho}(\pi_i; \pi_{-i}) \leq C_{\mathcal{G}} \max_{\tilde{\pi}_i} \langle \nabla_i V_i, \tilde{\pi}_i - \pi_i \rangle$
- SOS condition: $(\pi - \pi^*)^\top \mathrm{Jac}_v(\pi^*)(\pi - \pi^*) < 0$

5.1 Where the Ω-Framework is Superior

Dimension	Giannou et al.	Ω-Framework
Gradient structure	Single gradient $v(\pi) = \nabla_i V_i$.	Five-component gradient (eq. 27 of theos): exploration + exploitation + evidence-seeking + alignment + opponent-shaping.
Representation	Fixed tabular parameterization $\pi_i : \mathcal{S} \to \Delta(\mathcal{A}_i)$.	Learnable embedding ϕ_ψ with its own gradient (eq. **??**). Creates a bilevel optimization that Giannou doesn't address.
Loss decomposition	Single value function $V_{i,\rho}(\pi)$.	Trinity decomposition: $R = P_W \mathbb{E}[\mathcal{L} \mid W] + P_S \mathbb{E}[\mathcal{L} \mid S]$. Convergence can be tracked *separately* on truths vs. falsehoods.
GDP refinement	Single GDP inequality (their Lemma 2).	Trinity-refined GDP with separate W and S bounds. Rates may differ: the learner converges faster on truths than falsehoods (or vice versa).
Irreducible error	Not modeled — assumes the game is fully specified.	Explicit irreducible loss from the O-set (Pirsig's Quality, Gödel's incompleteness). The gap $Y - \hat{Y}$ has a floor that no gradient can eliminate.
Multi-agent structure	Agents are symmetric (same "type" of gradient).	Gödelian complementarity: agents with different F_i have structurally different blind spots B_i. Multi-agent > single-agent not by speed but by *coverage*.
Evidence quality	Not modeled.	Keynesian weight of evidence $V_i(\mathcal{X})$ as second loss axis. Evidence-seeking gradient term (eq. 11 of theos): $-\mu V^{-2} \nabla_\theta V$.
Convergence type	Local convergence to Nash with probability $1 - \delta$ (their Thm 1).	Same guarantee applies to the exploitation term. But the exploration term (REINFORCE) enables *global* search over O.

5.2 Formal Refinement: Trinity GDP

Proposition 1 (Trinity-Refined Gradient Dominance)**.** *Under the Ω-framework, for any policy profile $\pi = (\pi_i)_{i\in\mathcal{N}}$ and unilateral deviation π_i':*

$$V_{i,\rho}(\pi_i'; \pi_{-i}) - V_{i,\rho}(\pi_i; \pi_{-i}) \leq C_{\mathcal{G}} \Big[P_W \max_{\tilde{\pi}_i} \langle \nabla_i V_i^W(\pi), \tilde{\pi}_i - \pi_i \rangle + P_S \max_{\tilde{\pi}_i} \langle \nabla_i V_i^S(\pi), \tilde{\pi}_i - \pi_i \rangle \Big] \quad (14)$$

where V_i^W, V_i^S are the value functions restricted to states reachable through W- and S-propositions respectively.

Proof sketch. Apply Giannou's Lemma 2 separately to the W and S components of the reward, using the Trinity decomposition of the expected trajectory reward:

$$V_{i,s}(\pi) = \mathbb{E}\Big[\sum_{t=0}^{T(\tau)} R_i(s_t, \alpha_t) \Big| s_0 = s \Big] = \underbrace{\mathbb{E}\big[R_i \cdot \mathbf{1}_{X_t \in W}\big]}_{V_i^W} + \underbrace{\mathbb{E}\big[R_i \cdot \mathbf{1}_{X_t \in S}\big]}_{V_i^S}.$$

The GDP (which follows from the linearity of V in π_i) applies to each component since $V_i^W + V_i^S = V_i$. □

Corollary 2 (Asymmetric Convergence Rates)**.** *If π^* is SOS and the Trinity-refined GDP holds, then the convergence rate from Giannou's Theorem 2 applies separately to V_i^W and V_i^S, potentially with different constants μ_W, μ_S in the SOS condition. In particular, a learner may:*

- *Converge at rate $\mathcal{O}(1/n)$ on W-propositions (truths it gets right),*
- *Converge at rate $\mathcal{O}(1/\sqrt{n})$ on S-propositions (falsehoods it gets wrong),*

and the overall rate is dominated by the slower component.

5.3 The Five-Component Gradient vs. Giannou's PG Template

Giannou's PG template (their eq. PG):

$$\pi_{n+1} = \mathrm{proj}_{\Pi}(\pi_n + \gamma_n \hat{v}_n)$$

with $\hat{v}_n = v(\pi_n) + U_n + b_n$ (true gradient + noise + bias).

In the Ω-framework (Item 52), $\hat{v}_n$ decomposes as:

$$\hat{v}_{i,n} = \underbrace{\omega_i \mathbb{E}[\mathcal{L} \cdot s_{\theta_i}]}_{\text{(a) exploration}} + \underbrace{\omega_i \mathbb{E}[\nabla_{\theta_i} \mathcal{L}]}_{\text{(b) exploitation}} + \underbrace{\omega_i \mu(-V_i^{-2} \nabla_{\theta_i} V_i)}_{\text{(c) evidence}} + \underbrace{\lambda \nabla_{\theta_i} D}_{\text{(d) alignment}} + \underbrace{\sum_{j \neq i} \frac{\partial R_i}{\partial \theta_j} \frac{\partial \theta_j'}{\partial \theta_i}}_{\text{(e) opponent shaping}} \quad (15)$$

What Giannou captures: Only term (b), the exploitation gradient. Their $v(\pi_n)$ is the exact policy gradient, which is the exploitation term.

What Giannou misses:

- **(a) Exploration**: The REINFORCE score-function term that adjusts *where* the agent looks. This is not a noise term (U_n) — it's a systematic gradient component.
- **(c) Evidence**: The Keynesian evidence-seeking term. Absent from any standard MARL framework.
- **(d) Alignment**: The consensus penalty. Related to communication in Dec-POMDPs but not in Giannou's model.
- **(e) Opponent shaping**: The LOLA term. Giannou explicitly assumes agents treat others as stationary (their Model 1). The Ω-framework models opponent adaptation.

Remark 3. *Giannou's convergence results (Theorems 1–3) apply to the exploitation term (b) in isolation. The Ω-framework's contribution is showing that in practice, agents should optimize* all five terms simultaneously*, and the convergence guarantees from Giannou give a lower bound on what the exploitation term alone achieves.*

6 The SOS Condition and Trinity Curvature

Giannou's SOS condition (their eq. SOS):

$$(\pi - \pi^*)^\top \mathrm{Jac}_v(\pi^*)(\pi - \pi^*) < 0 \quad \text{for all } \pi \in \Pi \setminus \{\pi^*\}.$$

In the Ω-framework, the Jacobian Jac_v decomposes via the Trinity:

$$\mathrm{Jac}_v(\pi^*) = P_W \cdot \mathrm{Jac}_{v^W}(\pi^*) + P_S \cdot \mathrm{Jac}_{v^S}(\pi^*). \tag{16}$$

This means the SOS condition can be *refined*:

Definition 4 (Trinity-SOS)**.** *A Nash policy π^* is **Trinity-SOS** if:*

$$(\pi - \pi^*)^\top Jac_{v^W}(\pi^*)(\pi - \pi^*) < 0 \quad \textit{(SOS on truths)} \tag{17}$$

$$(\pi - \pi^*)^\top Jac_{v^S}(\pi^*)(\pi - \pi^*) < 0 \quad \textit{(SOS on falsehoods)} \tag{18}$$

for all $\pi \neq \pi^*$ sufficiently close.

Remark 5. *Trinity-SOS is* stronger *than Giannou's SOS (it implies SOS by linearity) but gives* finer *convergence information. The eigenvalues of Jac_{v^W} and Jac_{v^S} independently control convergence on truths vs. falsehoods. A policy might be SOS overall but unstable on S (the falsehood component has a positive eigenvalue), meaning the learner is systematically worsening on falsehoods while improving on truths — a phenomenon Giannou's framework cannot detect.*

7 Summary: What the Ω-Framework Adds

1. **The gradient is richer**: Five components vs. one. Giannou's convergence results give a floor for what exploitation alone achieves; the Ω-framework shows the full gradient agents should compute.

2. **The loss decomposes**: The Trinity W/S split gives separate convergence tracking on truths vs. falsehoods. This is invisible in Giannou's framework.

3. **There's an irreducible floor**: The O-set (Gödel, Pirsig, Keynes) means $R(\theta^*) > 0$ for any θ^*. Giannou's framework implicitly assumes the game is fully specified; the Ω-framework models what is *structurally unknowable*.

4. **Representation learning matters**: The ψ-gradient (eq. **??**) with its inaccessible term $\partial f/\partial \mathbf{X}$ is absent from Giannou's tabular setting. This is where Pearl's causal structure (Item 46–48) provides the missing $\partial f/\partial \mathbf{X}$ via DAG-structured backpropagation.

5. **Multi-agent is structurally different**: Gödelian complementarity means multi-agent systems can know strictly more than any single agent — not by being faster (Giannou's concern) but by having non-overlapping blind spots. The five-component gradient (eq. **??**) captures this via the opponent-shaping term.

6. **The Taylor expansion is the entry method**: Equation (**??**) is the key decomposition. At any stationary point, the risk separates into irreducible loss (row 1), bias (row 2), and variance (row 3), each further split over W and S. This is the diagnostic tool: which component of the loss, on which partition of reality, is the learner failing?

Appendix: Lean 4 Formalization Summary

A Lean 4 file `Theos.lean` accompanies this document, formalizing:

- The truth value type `TruthVal` and partition W, O, S (Items 2–4)
- The Trinity exhaustiveness and disjointness theorems (proved)
- $O = \Pi^c$ (proved)
- The bijection $\tau : O \to \Pi$ (axiomatized)
- The UPRT: $\Pi \leftrightarrow \mathbb{R}^\infty$ (axiomatized)
- The learning pipeline: `LossFunction`, `Embedding`, `CausalMap`, `Approximator`, `LearningPipeline` (defined as structures)
- The gradient decompositions (axiomatized as structural claims)
- Connection to Giannou: the Ω-gradient subsumes Giannou's PG template (stated)

The file requires Lean 4 with Mathlib. Proofs marked `sorry` require analysis-level Mathlib (continuous differentiability, Fréchet derivatives) which is available but verbose to formalize fully.